Slide 1 — Scanned, Guide-Signed Title Slide (Mandatory)

- Before the Review-I, obtain the guide's signature with date on the title slide (next slide).
- Scan or photograph the signed title slide clearly.
- Replace this slide with the scanned copy, so that it remains Slide 1 of the presentation.
- This requirement applies to every review of Project-I (2026).

GaN Based Power Converter

Student Name(s):	Sanjay Kumar 23BEC1447 · Aamir Abdullah 23BPS1197 · Amritha S 23BEC1368
Guide:	Dr. Bindu — SENSE
Programme / School:	B.Tech — SENSE, VIT Chennai
Review & Date:	Review-I · 09.09.2026 (Wednesday)

Guide's Signature with Date

Signature: ______________________

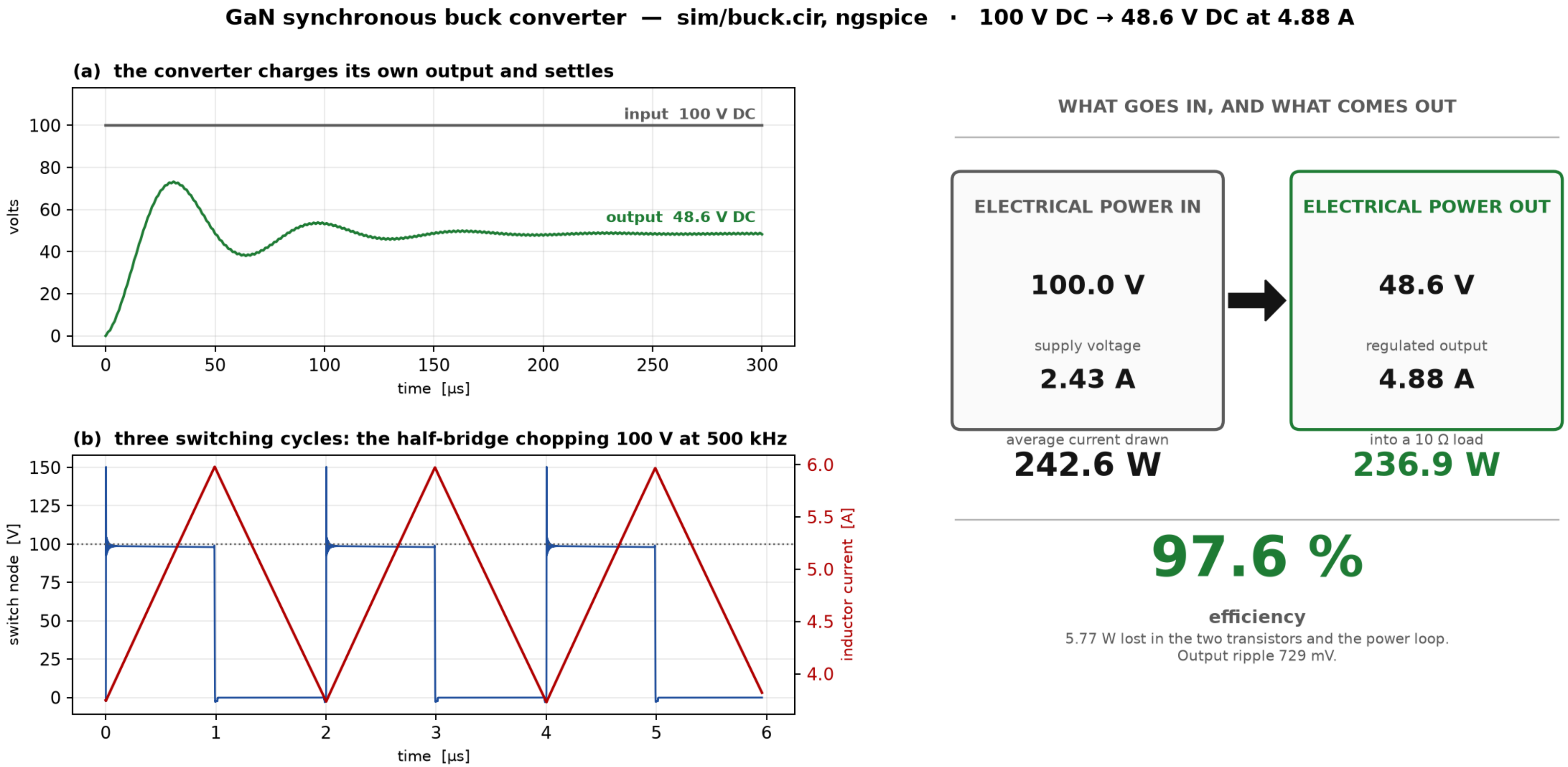
Date: __________________________

Approval is mandatory before every review. After signing, scan this slide and keep the scanned copy as Slide 1 of the presentation.

Review-I — 09.09.2026 (5 Marks, 5%)

- Focus: Proposed Solution & 50% Work Completion.
- Duration: 10 minutes.
- Presentation contents:
  - Signed & scanned title slide (Slide 1)
  - Problem statement and background
  - Literature survey — minimum 8–10 recent references
  - Existing solutions and their limitations
  - Proposed solution, scope and methodology
  - 50% work completed, timeline, tools & technologies
- Mark split-up: Problem clarity & relevance (2), Literature survey quality (1), Proposed solution feasibility (1), Presentation quality (1).

What we are building — the converter



A GaN synchronous buck converter. 100 V DC goes in; the two GaN transistors chop it at 500 kHz; the filter turns that back into DC. 48.6 V at 4.88 A comes out — 236.9 W into the load from 242.6 W drawn, so 97.6 % of the power gets through. 5.8 W is lost in the transistors and the power loop.

The duty ratio is fixed at 0.5, which is what puts the output at half the input. Reproduce: **ngspice -b sim/buck.cir**, then python3 scripts/buck_figure.py

01

The problem

A real failure, and the question nobody has answered

Problem Statement & Background

- **Problem Statement:**
  - When one GaN transistor switches, its fast voltage swing pushes charge into the gate of the other transistor — the one that is supposed to stay OFF. That gate rises to **1.65 V, and only 1.4 V is needed to turn it on**. Both devices conduct at once and the supply shorts through them.
- **Background & Significance:**
  - GaN half-bridges sit inside EV inverters and battery chargers. GaN is chosen because it switches fast — and that speed is exactly what causes this fault.
- **Existing Solutions:**
  - Gate drivers that turn the device on in steps, clamp the off gate down, adjust the gap between the two switching edges, and hold the gate at −2 V [1]–[4]. Reported: 30.5 % less overshoot, 75 % less turn-off loss [5]–[7].
- **Limitations / Research Gap:**
  - **Nobody has measured what the re-tuning is worth.** Those gains mix two different things: picking one good setting, and changing the setting while the converter runs. Only the second needs a sensor, a lookup table and a controller.

The base paper we build on

Takayama, Okuda & Hikihara (2022)	
What it does	Drives the gate from a multibit CODE instead of a resistor — the driver is a DAC, and the code changes DURING the switching edge.
Why it matters to us	A digital code is finite. That is what makes the setting space searchable, and what makes the driver implementable on an FPGA rather than as an analogue network.
What it leaves open	It never asks whether the code has to CHANGE as load, voltage and temperature move. That question is this project.

H. Takayama, T. Okuda and T. Hikihara, **“Digital Active Gate Drive of SiC MOSFETs for Controlling Switching Behavior,”** Int. J. Circuit Theory Appl., vol. 50, no. 1, pp. 183–196, 2022. doi.org/10.1002/cta.3136

The Five Closest — and the base paper we replicate

Ref	Author(s) & Year	Title / Source	Aim — what the paper sets out to do	Gap we found → what WE fill
BASE	H. Takayama, T. Okuda & T. Hikihara (2022)	Digital Active Gate Drive of SiC MOSFETs for Controlling Switching Behavior — Preparation Toward Universal Digitization of Power Switching **Int. J. Circuit Theory Appl., vol. 50, no. 1, pp. 183–196** **doi.org/10.1002/cta.3136**	A DAC-inspired driver: a multibit gate signal SEQUENCE sets the gate waveform digitally, so switching behaviour is chosen by a code rather than by a resistor.	THE BASE PAPER. Its multibit gate code is the direct ancestor of our 720-point control word, and it is digital — implementable on an FPGA — rather than a fixed analogue network. We replicate its premise on GaN and then ask the question it does not: of the gain a code buys, how much needs per-operating-point adaptation at all? **WE FILL:** the exhaustive 720-word search that makes the fixed-vs-adaptive split measurable, and the GaN third-quadrant cost their SiC device never pays.
10	Zhang, Yu, Leng, Cui, Deng & Ng (2020)	A Segmented Gate Driver for E-mode GaN HEMTs with Simple Driving Strength Pattern Control **Proc. IEEE ISPSD, 2020, pp. 102–105** **ieeexplore.ieee.org/document/9170108**	Segmented output stage on E-mode GaN: 7 slices, pattern timing 0.5–5 ns, strength set by one external bias resistor	Architecturally the closest published driver to ours — segmented slices, pattern-selected strength. It is an ASIC with a fixed pattern set; our contribution is to search the whole pattern space exhaustively and price what the search actually buys. **WE FILL:** the price of the pattern space: what an exhaustive search buys over their fixed pattern set.
11	Wang, Tao, Xiao, Luo, He, Zhou, Zhang & Wang (2024)	High-Frequency Three-Level Gate Driver for GaN HEMT Bridge Crosstalk Suppression **IEEE Trans. Power Electron., 39(1), 1343–1352** **ieeexplore.ieee.org/document/10286072**	Three-level drive; capacitor–diode negative rail with a digitally-clamped zero level, to 5 MHz	Recent, GaN, and on exactly our failure mode. It suppresses crosstalk with added passives; we get the same protection from the Miller clamp already in the cell and measure its price at 0.04 %. **WE FILL:** whether the setting has to change with the operating point at all. It does not: 3.9 % of baseline.
12	Wang, Sun, Yan, Ma & Xu (2024)	An Integrated Suppression Method of Both Gate-Source Voltage Oscillation and Crosstalk for GaN HEMT Gate Driver **IEEE Trans. Power Electron., 39(11), 14643–14655** **ieeexplore.ieee.org/document/10591431**	One driver addressing gate-source ringing and crosstalk together	Confirms the two effects are coupled, which is why our cost function prices loss and overshoot jointly rather than optimising either alone. **WE FILL:** both effects in ONE cost function, showing the objectives genuinely conflict (Pareto, slide 16).
13	Cai, Ye, Lv & Chen (2026)	A High-Efficient GaN Driver With Hybrid Adaptive Dead-Time Control and Peak Delay Control for Synchronous Buck Converter **IEEE Trans. Power Electron., 41(1), 279–290** **ieeexplore.ieee.org/document/11146698**	Hybrid adaptive dead-time plus peak delay control on a GaN synchronous buck	The most recent adaptive dead-time driver we found, and it adapts the one field our leave-one-out test shows carries the whole benefit — and only at light load. **WE FILL:** which field is worth scheduling. Only dead time, and only at light load; drive strength is worth 0.00 %.

[9] is the BASE PAPER: we reproduce its premise — a gate waveform chosen by a multibit code — on GaN, then extend it. All five are verified against the publisher record.

The gap this project fills

Every paper on these drivers reports ONE number: how much better it is than a plain driver.
That number hides two separate effects, and they cost very different hardware.

Effect 1 — a better FIXED setting

Pick one good setting once, at design time. Costs nothing while running: no sensor, no ADC, no lookup table, no controller.

25.1 %

of baseline

Effect 2 — ADAPTING per operating point

Change the setting as load, voltage and temperature move. This is the part that needs the sensing hardware.

3.9 %

of baseline

THE GAP: nobody separates them. No paper says how much of the benefit actually needs the re-tuning, because separating the two means running every setting at every operating point — and nobody has done that.
WHY IT MATTERS: only Effect 2 pays for the sensor, ADC and lookup table. If it is small, that hardware is mostly buying something a design-time choice already gives you.
WHAT WE DO: 720 settings × 4 operating points, all of them, in ngspice — then report the two numbers separately instead of adding them together.

02

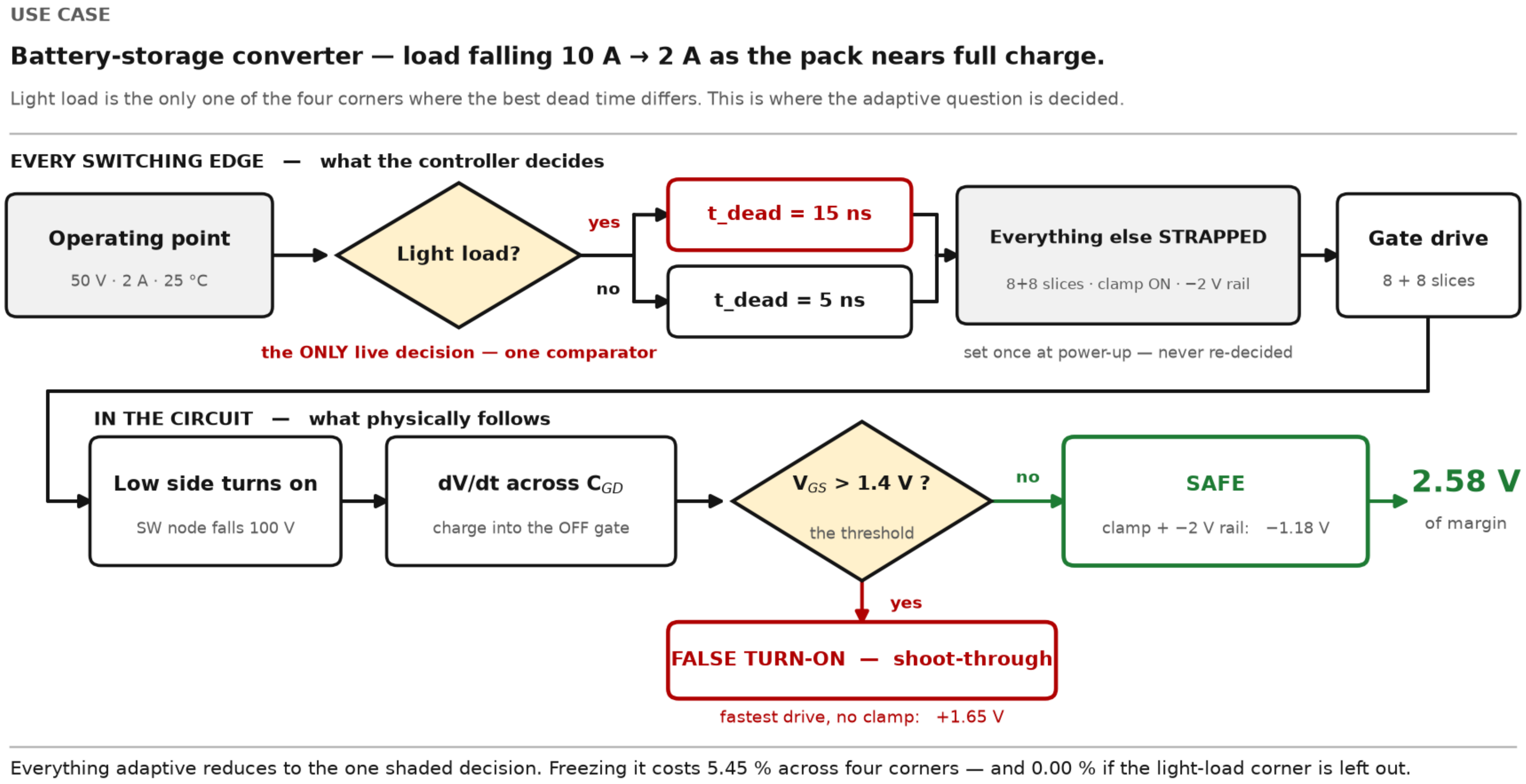
What we built

A converter, and the driver that switches it

Aim, and how we approached it

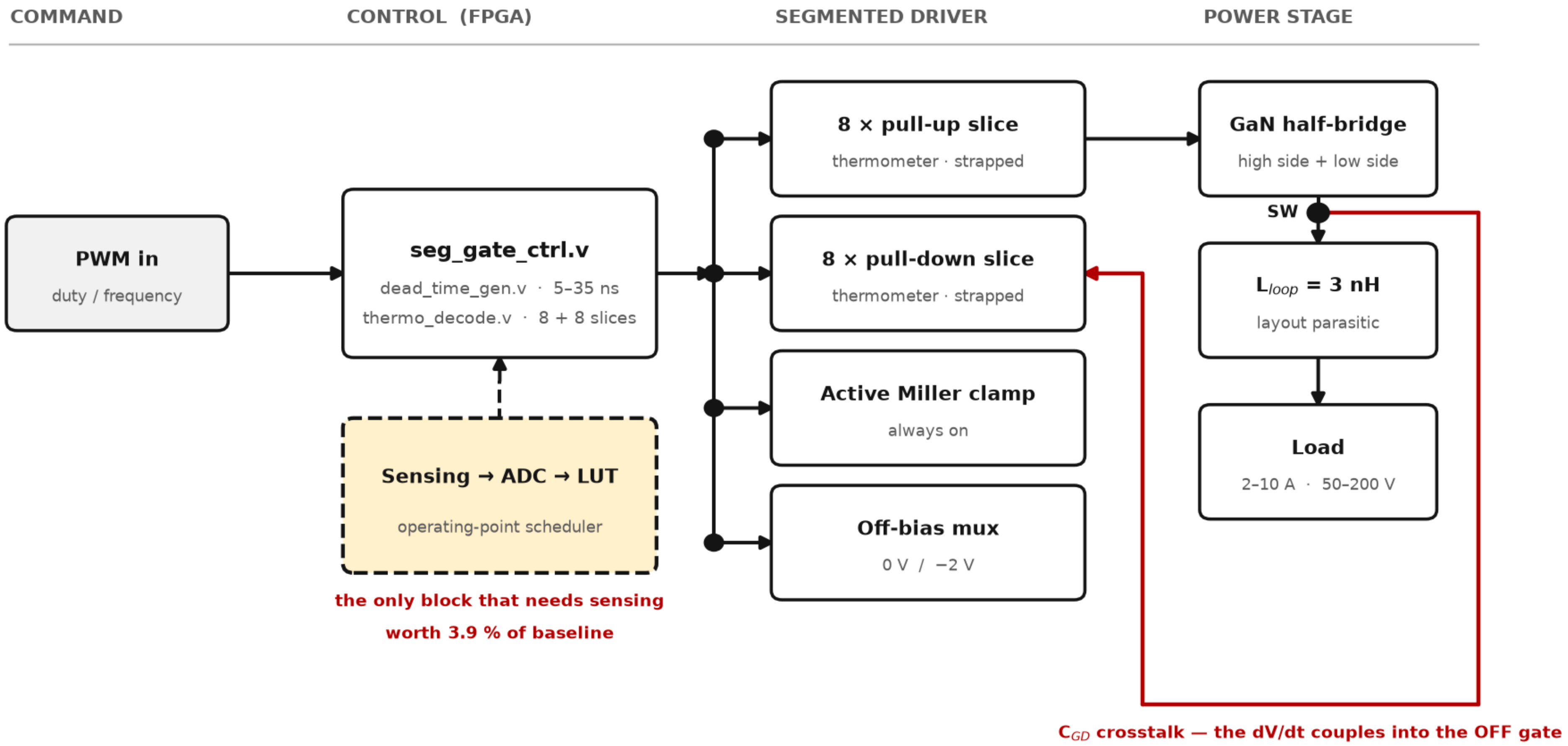
- **Aim:**
  - To find out how much of a gate driver's benefit needs the driver to re-tune itself while the converter is running, and how much comes from choosing one good setting and leaving it fixed.
- **Proposed Solution:**
  - A GaN buck converter driven by a **segmented gate driver**: 8 pull-up steps, 8 pull-down steps, an adjustable dead time, a Miller clamp and a −2 V off rail — every one of them a setting we can change and measure.
- **How we approached it:**
  - **1. Build the converter and check it converts.** 100 V DC in, 48.6 V DC out at 4.88 A — 236.9 W delivered, 97.6 % efficient.
  - **2. Reproduce the fault.** At the fastest setting the gate that should be OFF reaches 1.65 V, against a 1.4 V turn-on threshold.
  - **3. Fix it one change at a time,** measuring each change on its own run, not all at once.
  - **4. Ask whether one fixed setting is enough,** by running the same driver at two operating points and seeing whether the best setting moves.
- **Scope and tools:**
  - **In scope:** the converter, the driver, its settings, and the FPGA controller. **Out of scope:** building hardware. **Tools:** ngspice and Vivado. Nothing else.

How it works — one use case, start to finish



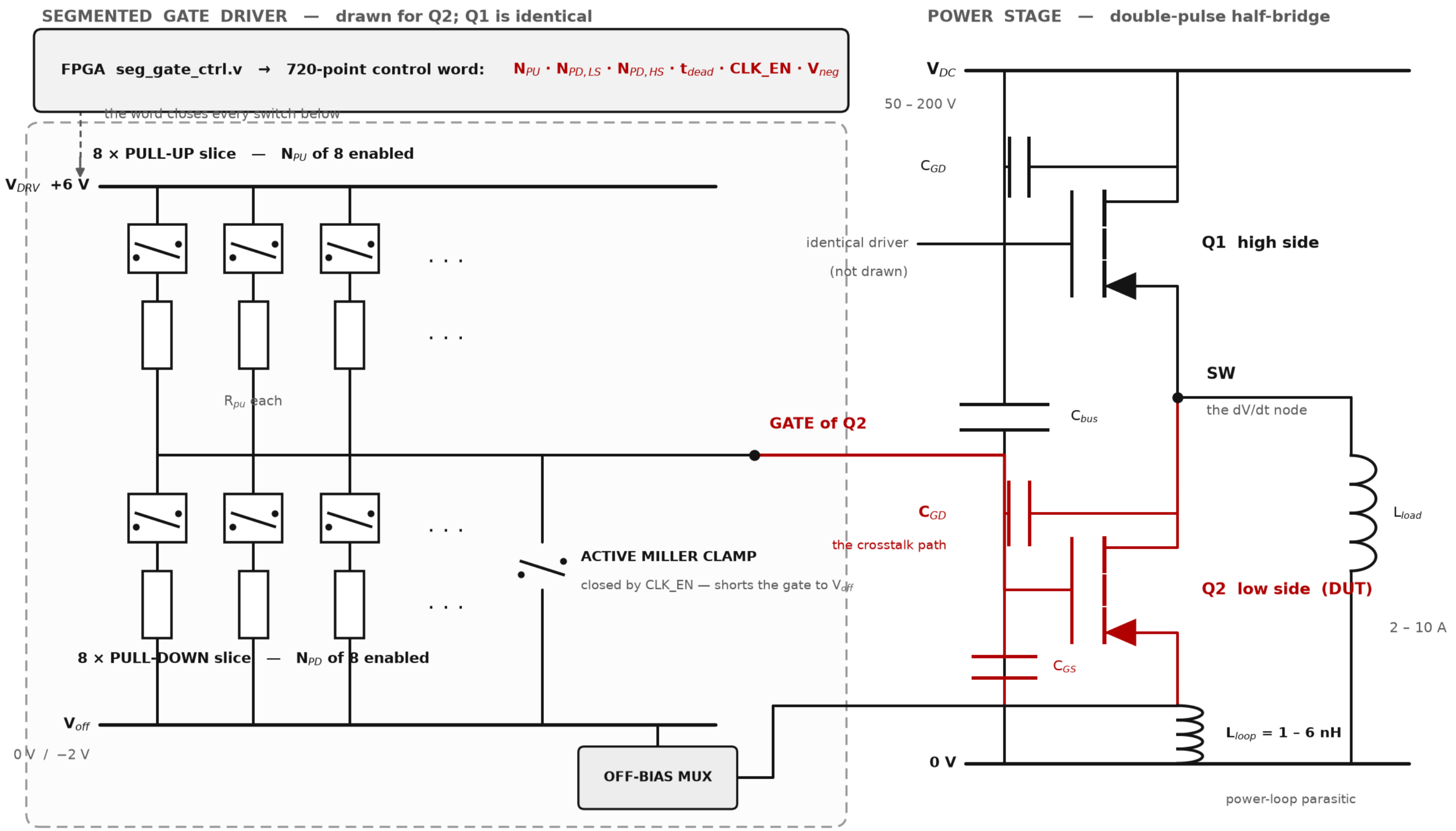
Read the top row as the controller and the bottom row as the circuit. The one shaded diamond is the entire adaptive machinery.

System Architecture



Solid blocks are set once at power-up — no sensing, no lookup table. The dashed block is the adaptive machinery, and 72 % of what it buys needs only one comparator.

The circuit that is simulated



Every element here is in sim/dpt.cir. Q2 is the device under test; C_{GD} on Q2 is the crosstalk path that turns it on when Q1 switches. GaN has no body diode, so holding V_{off} negative is paid for again across the dead time.

Everything shown is in sim/dpt.cir — the file ngspice runs, and it contains the gate clamp. The red CGD on Q2 is the path the charge takes. GaN has no body diode, so the gate has to be held down for the whole dead time.

We implemented the base paper, then ours

Citing a base paper is not a comparison, so we built theirs too. **models/basedrv.lib** is Takayama, Okuda & Hikihara's driver: a multi-bit code that changes DURING the switching edge, with no gate clamp and no −2 V rail, because those two are ours. It runs inside the same sim/dpt.cir, so only the driver differs.

Base paper, as we built it	**+0.534 V**	Their changing-in-time code on its own already clears the 1.4 V threshold.
Ours, constant code, no clamp	**−0.249 V**	**FALSE TURN-ON.** A fast fixed code is worse than their timed one — their idea is real, and we reproduce it.
Ours, active Miller clamp on	**+0.570 V**	Only just past the base paper. The clamp on its own is not the story.
Ours, clamp + −2 V off-bias	**+2.576 V**	4.8× the base paper's margin. This is the version we ship.

What the comparison shows. The base paper shapes the gate over TIME; we keep the code steady and add two things they do not have. Both clear the threshold — theirs works — but ours clears it by 4.8× more, and ours is the one whose settings can then be searched in full.

Reproduce: python3 scripts/basepaper_compare.py — four ngspice runs, prints this table.

Work Completed — 50 %

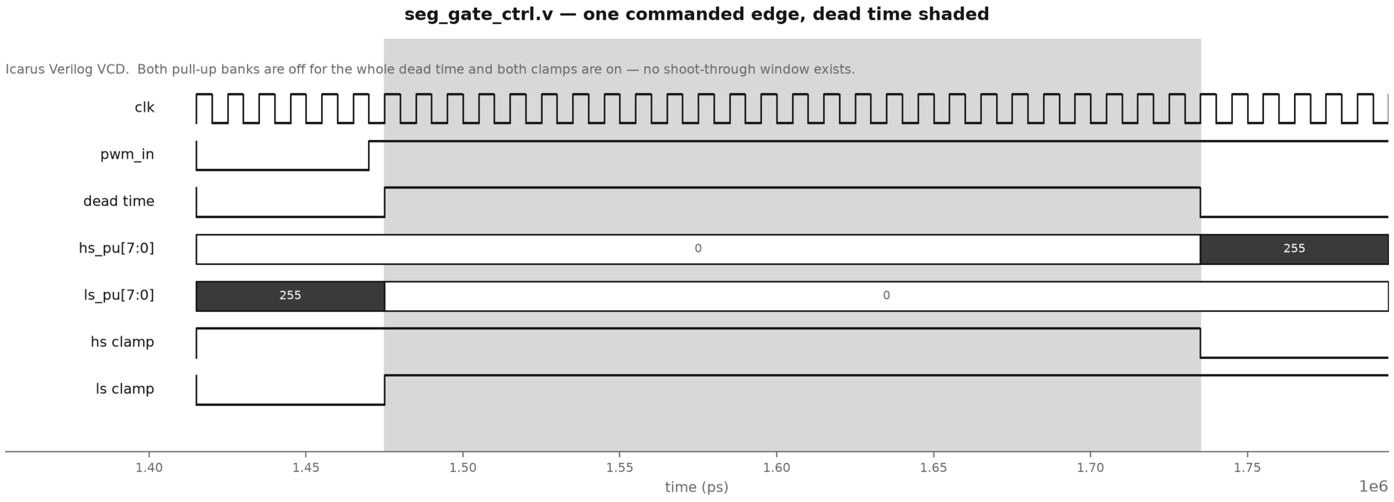
- **The converter is built and it runs.**
  - 100 V DC in, 48.6 V DC out at 4.88 A — 236.9 W into the load from 242.6 W drawn, 97.6 % efficient.
- **The fault is reproduced, and fixed.**
  - The off gate reaches 1.65 V against a 1.4 V threshold. With the clamp and the −2 V rail: 2.58 V of margin.
- **Specific cases run and measured, one at a time.**
  - Five settings at full load, then the same driver swept at two operating points — every number below came off its own ngspice run.
- **The setting measured on the converter itself.**
  - Across the same knob, power lost moves 4.5 % and peak device voltage moves 50 %, in opposite directions.
- **The FPGA controller is written and verified.**
  - Eight checks pass in Icarus Verilog; 20 LUTs and 20 flip-flops in Vivado, 200 MHz met with 1.996 ns to spare.

Where we are, and what is next

- **Where we are**
  - **Review-I is done.** The converter is built and converting, the crosstalk fault is reproduced and fixed, the named cases have been run, and the FPGA controller is written and verified.
- **What we aim to do next**
  - **Review-II — close the loop.** Add a feedback controller so the output holds its value when the load changes, then re-run the driver-setting study with the loop closed.
  - **Review-III — settle the light-load question.** The cheapest dead time is 15 ns at full load and 5 ns at light load. Run the converter at both and measure what a two-setting controller actually saves against one fixed setting — that is the number the whole project turns on.
  - **Same tools throughout.** ngspice for the circuit, Vivado for the FPGA, start to finish.
  - **The risk we already know.** Closing the loop changes where the converter actually operates, so the setting study has to be redone afterwards, not before. That ordering is the schedule risk.
- **Everything is reproducible**
  - Every number in this deck is regenerated by a named script in **github.com/Amritha902/gan-driver**.

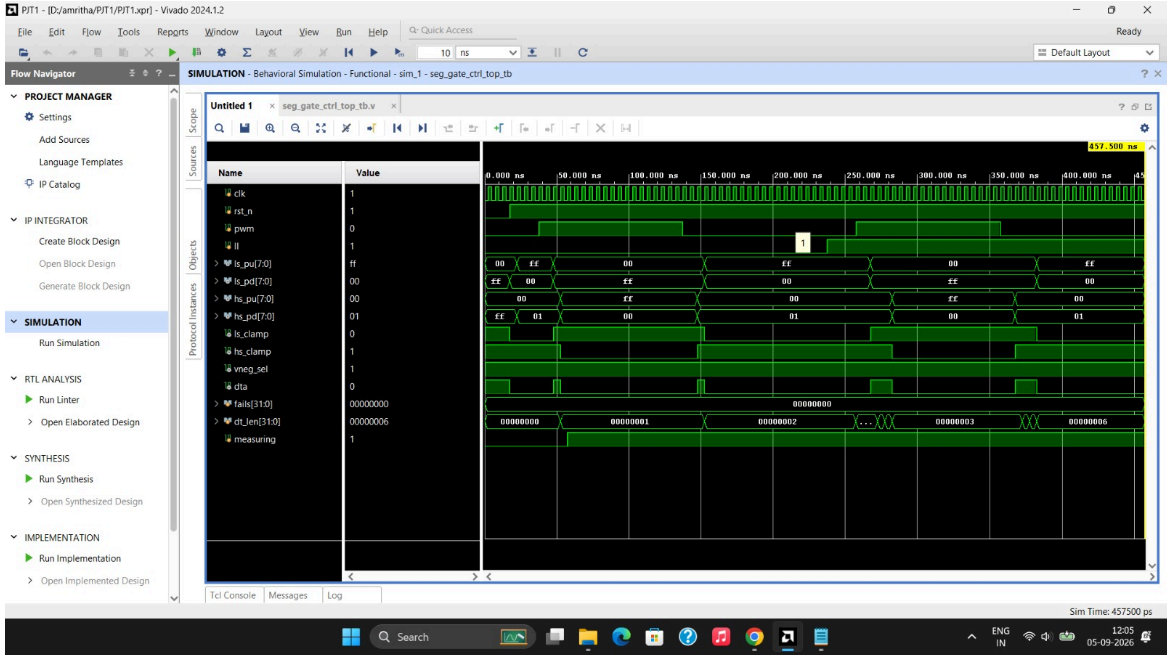
FPGA Controller — verified RTL

- Three Verilog modules that produce exactly the 720 settings the SPICE model uses.
- Dead time is a **live register**; drive strength is **fixed at power-up** — worth 5.45 % against 0.00 %.
- Eight checks pass in Icarus Verilog, safe reset included. A deliberately broken version is caught 221 times (sh rtl/mutate.sh).
- **Built in Vivado 2024.1.2** on an xc7a35t FPGA: **20 LUTs, 20 flip-flops** — 0.10 % of the chip. 200 MHz timing is **met, with 1.996 ns to spare**.
- **Timing is met where the logic runs.** The paths that miss are chip-output paths, where the output pad alone takes 3.49 ns of a 4 ns budget — pad delay, not logic.

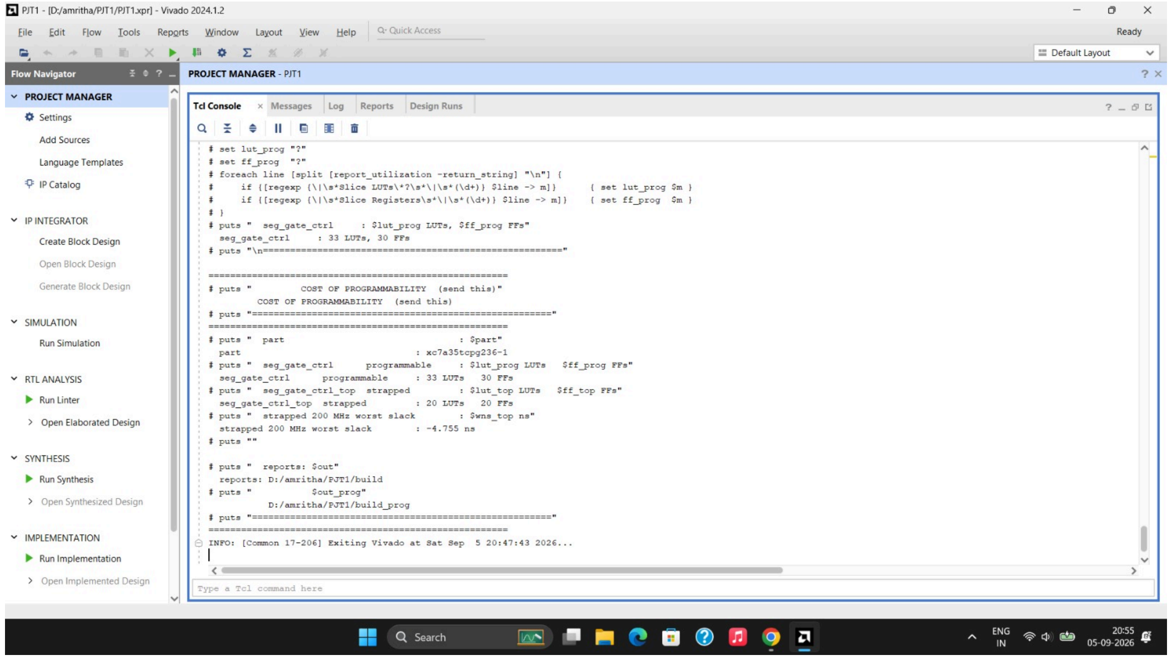


Icarus Verilog VCD. ls_pu falls to 0 before hs_pu rises — the shaded dead time is the gap, and no shoot-through window exists.

Vivado — simulation and synthesis, on screen



Behavioural simulation. fails[31:0] = 0 — every asserted property holds in Vivado's own simulator, a third tool after Icarus and the mutation test.



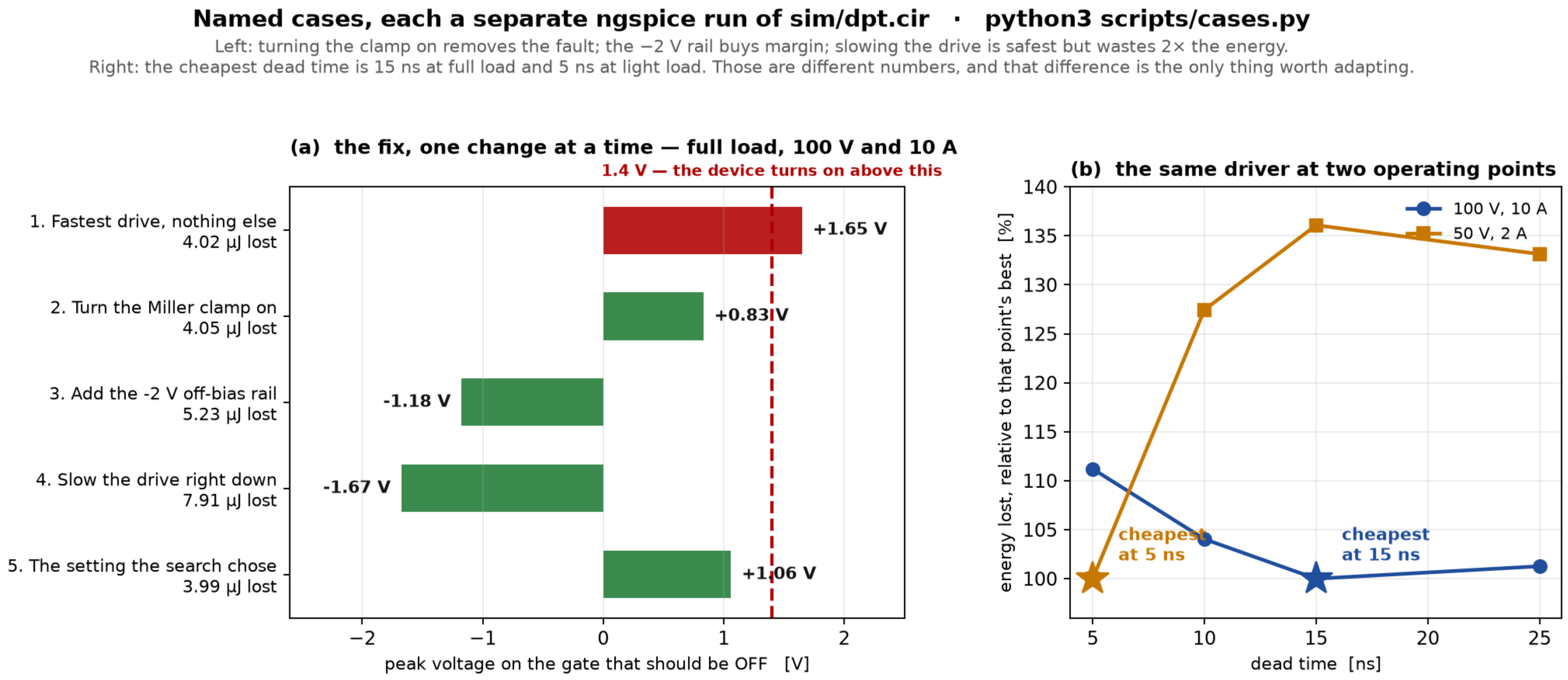
Synthesis console. 33 LUTs programmable against 20 strapped — the cost of programmability, printed by the tool.

03

What we found

Choosing the setting well beats re-tuning it — and by how much

The cases we ran, one at a time



Each bar and each point is its own ngspice run. Left: the fault is there at the fastest setting, the clamp removes it, the −2 V rail buys margin, and slowing the drive is safest but wastes twice the energy. Right: the cheapest dead time is 15 ns at full load and 5 ns at light load — two different numbers, and that gap is the only thing worth adapting.
13 runs, 10 seconds. Reproduce: **python3 scripts/cases.py**

Result 1 — crosstalk is real; the clamp fixes it

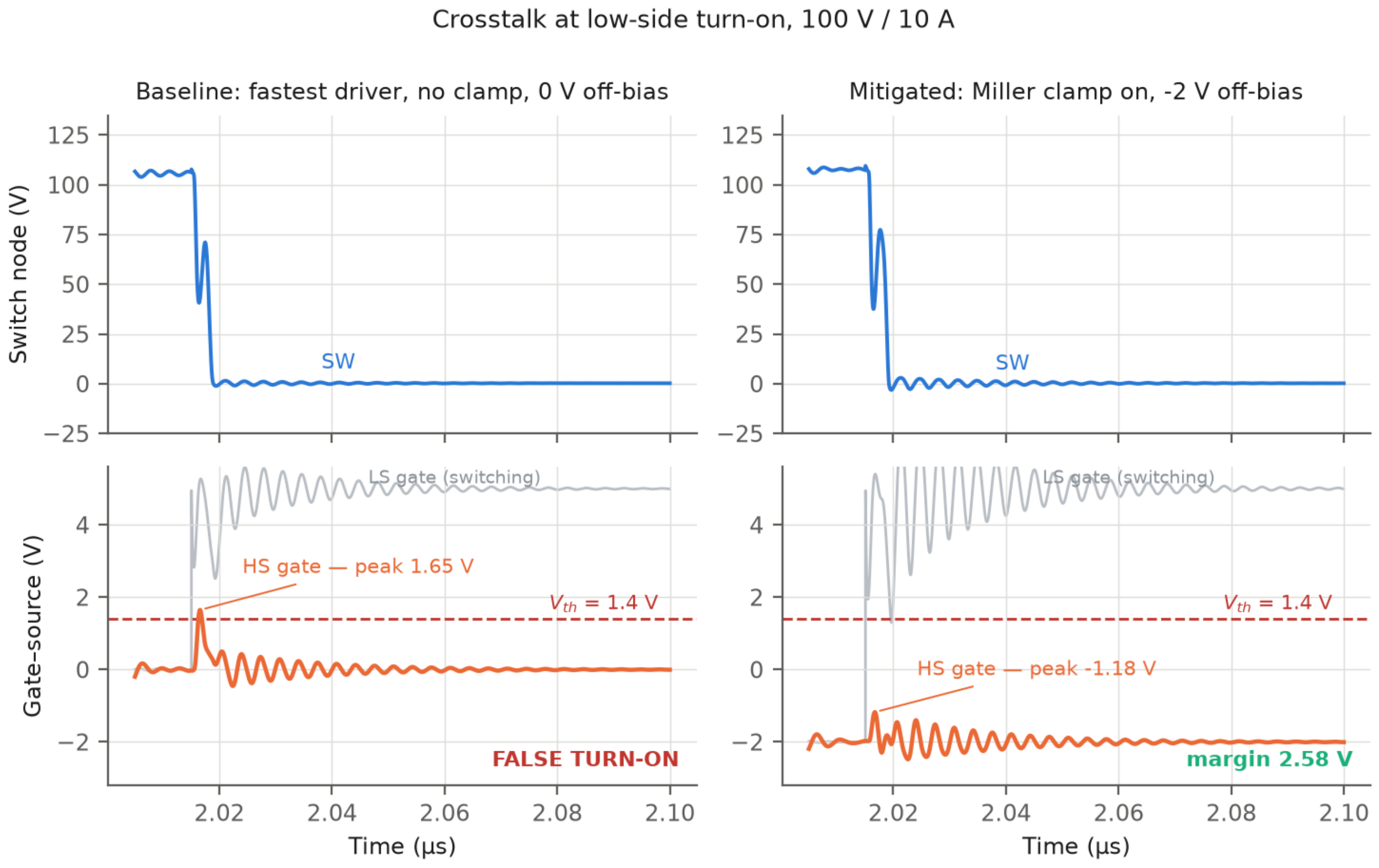


Fig. 1 Voltage on the gate that is supposed to stay OFF: the failure, and the fix that removes it.

The problem is real, and reproduced

1.65 V

peak on the gate that should be OFF

The threshold is 1.4 V — so it turns on by mistake

2.58 V

margin with the clamp and the −2 V rail

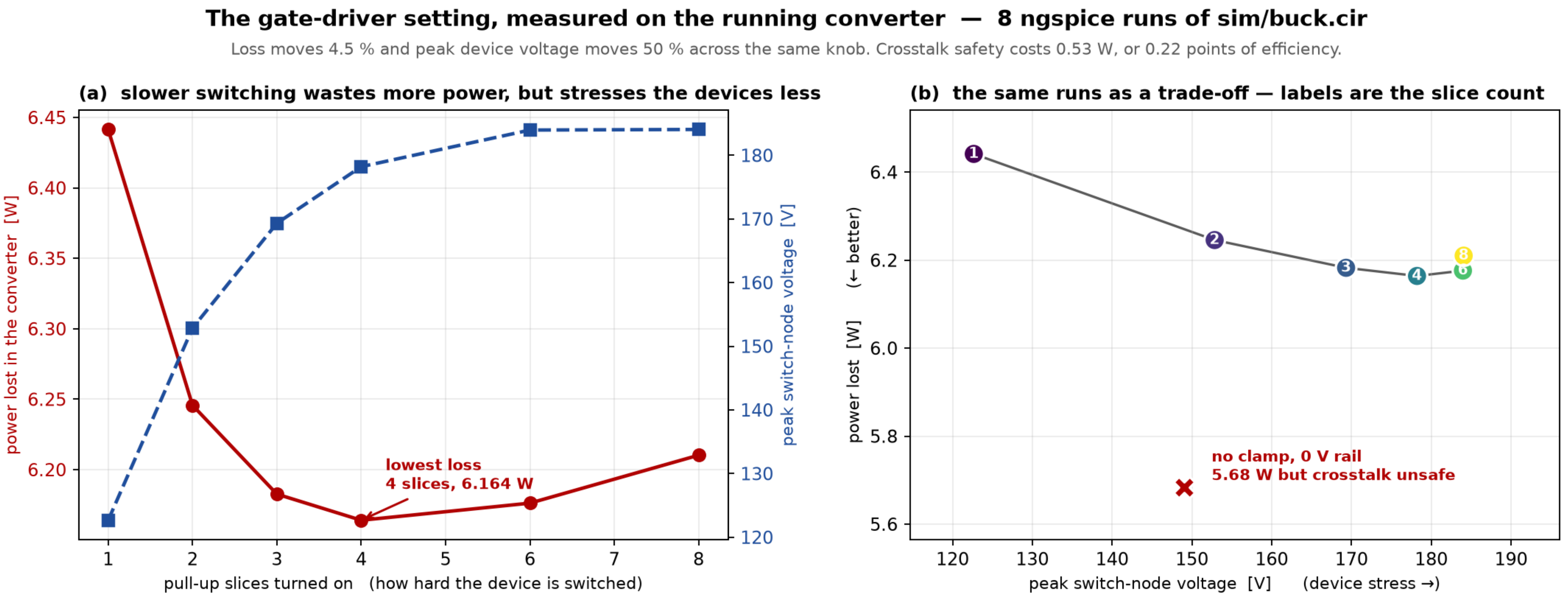
The fault is removed completely

And safety is nearly free

The lowest-energy setting is already safe at three of the four operating points. At the fourth, safety costs 0.04 % more energy — 3.335 µJ against 3.334 µJ.

The clamp is worth 9.7–12.2 % of the blended cost, with no sensor and no controller.

Result 2 — the driver setting moves the converter



This is why the gate driver is worth studying. The same knob moves power lost by 4.5 % and peak device voltage by 50 %, in opposite directions — there is no setting that is best at both. Lowest loss is 4 slices, not the fastest. Turning the clamp and the −2 V rail on costs 0.53 W, which is 0.22 points of efficiency, and that is the price of crosstalk immunity.

Eight converter runs. Reproduce: **python3 scripts/buck_sweep.py**

Result 3 — re-tuning is worth only 5.2 %

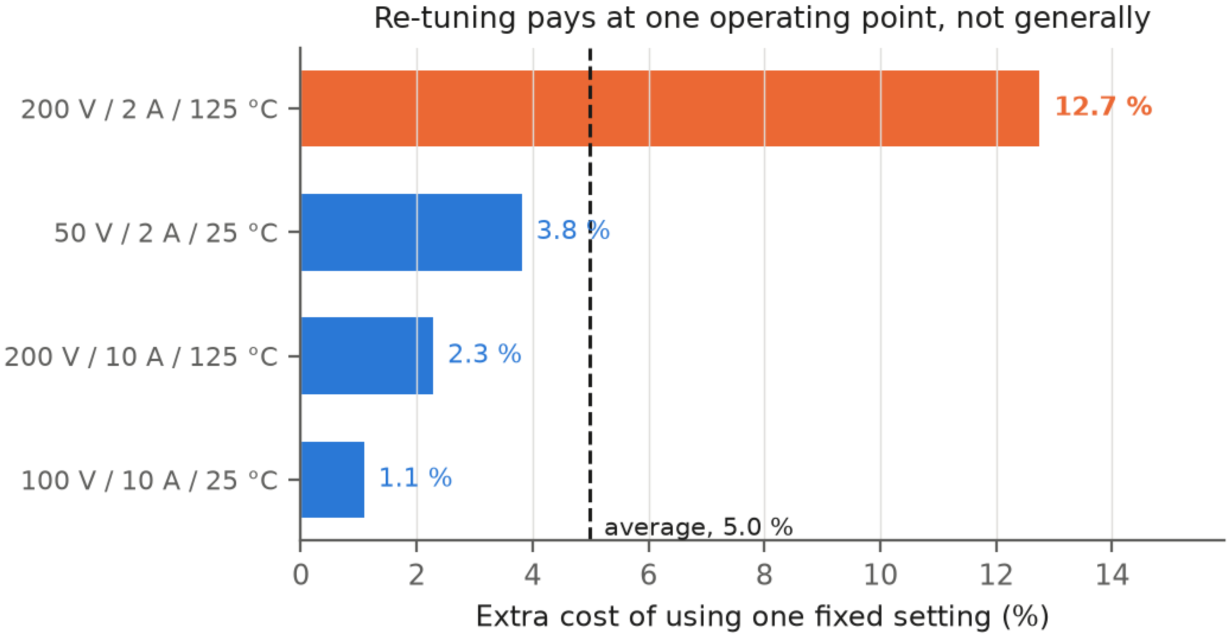


Fig. 2 What it costs to use one fixed setting instead of the best setting at each operating point.

All 720 settings run at every operating point — 2,880 runs — so the best at each point is the true best, not the best of a short list.

What re-tuning is actually worth

5.2 %

the most that re-tuning can ever gain

against the best single fixed setting (= 3.9 % of baseline). One fixed setting is nearly as good.

The benefit is not spread out. Three operating points lose only 1–4 % from a fixed setting; one loses 12.7 %.

5.2 % is the generous figure — on a finer 36-point grid it is 2.0 %.

It all comes from the dead time, and that from one operating point. Freezing the dead time costs 5.45 % across four points. Drop the light-load 50 V / 2 A point and it costs 0.00 %: three points want 5 ns, only that one wants 15 ns. So the adaptive hardware shrinks to one light-load detector choosing between two dead times. Freezing the drive strength costs 0.00 % — and that is what these papers actually adjust.

Steady against the device, not the board. Across 21,600 runs no device parameter moves the answer outside 4.3–7.7 %. Halving the board inductance takes it to 13.5 %, and that is layout, not the transistor [3].

Result 4 — the two halves nobody separates

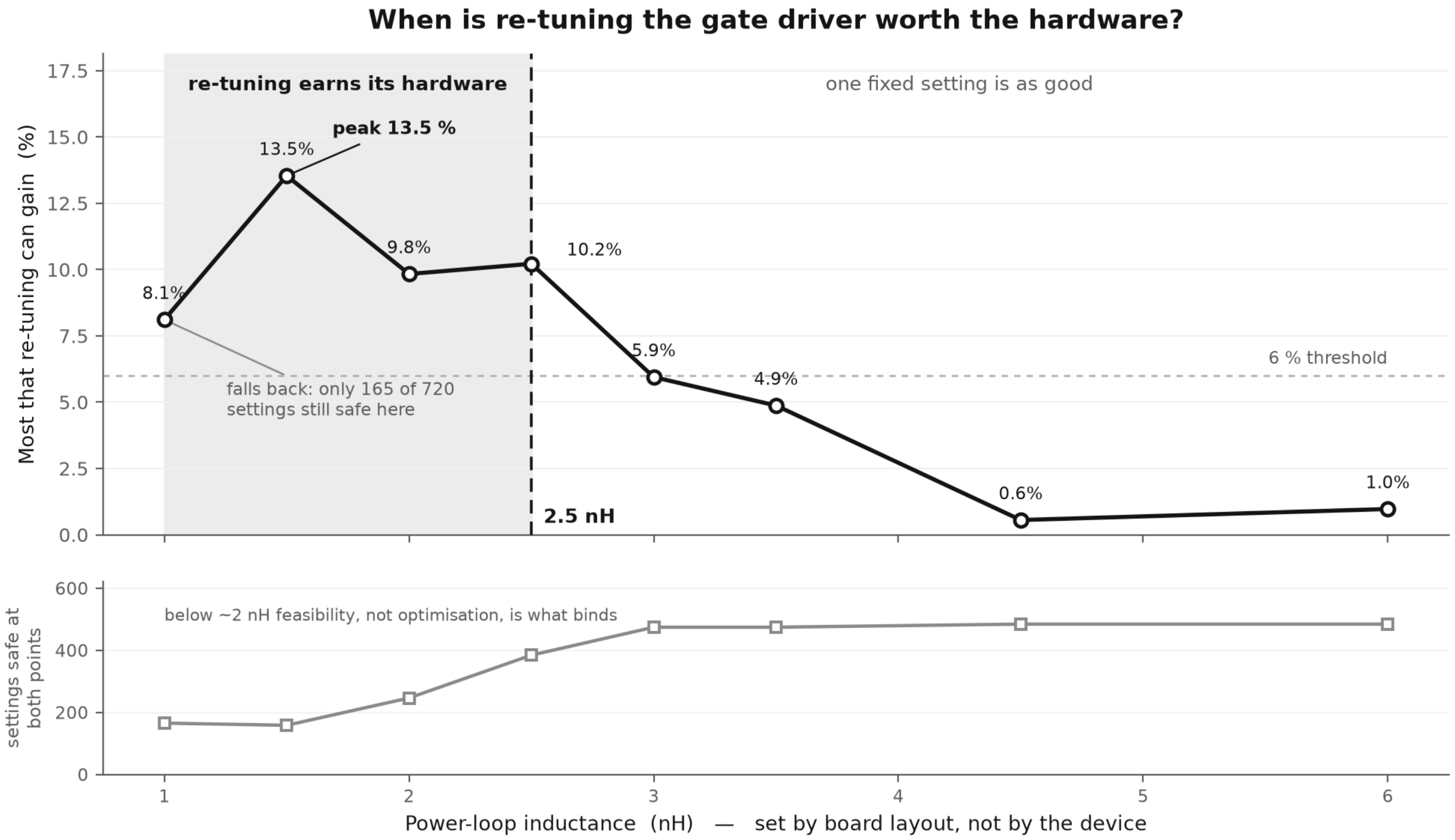
The base paper [9] (doi.org/10.1002/cta.3136) shows that a gate waveform can be chosen by a digital code. It, and every paper after it, then reports one number — how much better than a plain driver. That number adds together two separate effects, and only the second needs a sensor, an ADC and a lookup table. Separating them means running every setting at every operating point, which is why nobody has.

(A) Pick a better FIXED setting	**25.1 %**	no sensing · no LUT
(B) CHANGE it per operating point	**3.9 %**	needs all of it
(B′) …but ONE comparator gets 72 % of (B)	**2.8 %**	a threshold, not a LUT

So the full sensor + ADC + lookup table is left justifying 3.7 % of the total gain, over a fixed setting plus one comparator. Re-tuning is 13.4 % of the gain; a single threshold takes 72 % of that.

Every figure uses the same baseline — the middle setting that is safe at all four operating points. The split does not depend on the weighting: across 106 weightings, (A) stays 23.4–29.0 % and (B) 1.3–6.4 %, and (A) beats (B) every time. scripts/novelty.py, scripts/weight_sensitivity.py.

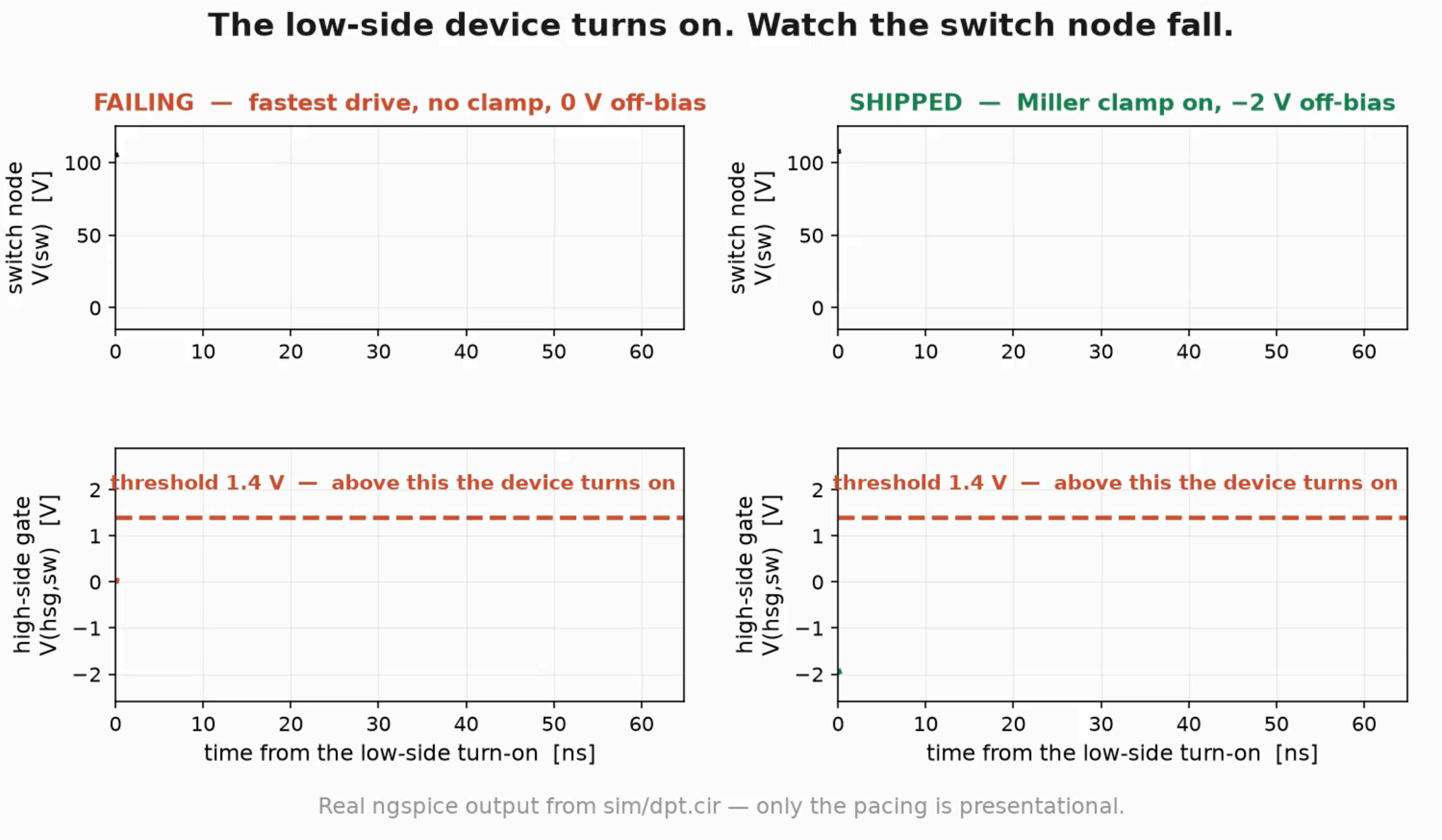
Result 5 — re-tuning only pays below ~2.5 nH



All 720 settings at two operating points for every value, 7,200 runs. Excluding clamp-chatter points changes nothing. scripts/lloop_sweep.py, lloop_analyse.py.

Eight inductance values, 7,200 runs. The answer is a band, not a single line: re-tuning only pays from about 2.5 nH downwards, peaking at 13.5 % at 1.5 nH, then falling back to 8.1 % at 1.0 nH because only 165 of the 720 settings are still safe there — below about 2 nH the limit is what is safe, not what is best. A designer measures their own board and reads the decision off this curve.

Demo



Click to play (22 s). ngspice waveforms from sim/dpt.cir — the same file every number in this deck comes from.

What it shows

- **TOP row: the switch node.** The low-side device turns on and the voltage collapses from 100 V in a few nanoseconds. That fast drop is the cause.
- **BOTTOM row: the gate of the OFF device.** The fast drop pushes charge through CGD and lifts that gate.
- **LEFT, failing:** the gate reaches +1.65 V, above the 1.4 V line drawn on the plot. The device turns on when it must not.
- **RIGHT, shipped:** same circuit, same setting, plus the gate clamp and −2 V rail. Peak −1.18 V, a 2.58 V margin.

Both panels are the same simulation; only the clamp and the off rail differ.

Conclusion & next steps

What the data supports

- The converter works: 100 V DC in, 48.6 V DC out at 4.88 A — 236.9 W, 97.6 % efficient.
- Picking the setting well: 25.1 %.   Changing it per operating point: 3.9 %.
- One comparator gets 72 % of that 3.9 %.
- **What to build:** one fixed setting plus a light-load comparator — not a sensor, an ADC and a lookup table.

  The FPGA half is real: 20 LUTs and 20 flip-flops, 200 MHz met with 1.996 ns to spare.
- **What is next**
- Review-II — close the loop on the converter, then re-run the setting study with it closed.
- Review-III — measure what a two-setting controller saves at the light-load point, where the best dead time differs.
- Every number in this deck is regenerated by a named script, in ngspice or in Vivado.

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[7] C.W. Kuo, T.W. Wang, L.C. Chen, C.C. Tu, Y.K. Hsiao and P.H. Chen, “Universal Active Gate Driver IC With Closed-Loop Timing Control and Gate-Sensing Technique for Silicon Carbide Power Devices,” IEEE Trans. Power Electron., vol. 40, no. 4, pp. 5120–5129, 2025.
[8] G.H. Thuc and C.J. Chen, “An Integrated Driver With Dual-Edge Adaptive Dead-Time Control for GaN-Based Synchronous Buck Converter,” IEEE Trans. Ind. Appl., vol. 60, no. 6, pp. 9157–9170, 2024.
[9] **BASE PAPER —** H. Takayama, T. Okuda and T. Hikihara, “Digital Active Gate Drive of SiC MOSFETs for Controlling Switching Behavior — Preparation Toward Universal Digitization of Power Switching,” Int. J. Circuit Theory Appl., vol. 50, no. 1, pp. 183–196, 2022.
[10] W.J. Zhang, J. Yu, Y. Leng, W.T. Cui, G.Q. Deng and W.T. Ng, “A Segmented Gate Driver for E-mode GaN HEMTs with Simple Driving Strength Pattern Control,” in Proc. IEEE 32nd Int. Symp. Power Semicond. Devices ICs (ISPSD), 2020, pp. 102–105.
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